

Academic Curriculum Vitae

Timothy Charles Andrews

September 16, 2026

Place of Birth: Croydon, England

Nationality: Dual citizen of New Zealand and the United Kingdom

New Zealand Permanent Residence: 28 Plympton St, New Plymouth, 4310

United States Current Residence: 1105 Nielsen Court, Apt 6, Ann Arbor, Michigan, 48105

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Research Interests

- Numerical modelling of geophysical fluid equations
- Development and testing of dynamical cores for Global Circulation Models (GCMs)
- Diffusion mechanisms in dynamical cores
- Analysis and comparison of timestepping methods
- The phase-averaged timestepping method for oscillatory PDEs

Academic Positions

Postdoctoral Research Fellow - Department of Climate and Space Science, University of Michigan, United States of America (October 2024 – October 2027)

Education

Doctor of Philosophy (PhD) in Mathematics (Geophysical Fluid Dynamics) — University of Exeter, United Kingdom (September 2020 – September 2024)

Bachelor of Engineering with First Class Honours, specialising in Engineering Science — University of Auckland, New Zealand (2016 – 2019)

Selected Presentations

- Poster presentation at EGU 2026: **Andrews, T.**, Jablonowski, C., Hughes, O., and Bendall, T.: New mountain-generated mesoscale flow test cases from DCMIP2025, EGU General Assembly 2026, Vienna, Austria, 3–8 May 2026, EGU26-2009, <https://doi.org/10.5194/egusphere-egu26-2009>, 2026.
- Poster presentation at EGU 2025: **Andrews, T.**, and Jablonowski, C.: New mountain-generated mesoscale flow test cases for dynamical cores. AGU Fall Meeting 2025, New Orleans, USA.
- Oral presentations at the 2025 and 2026 Community Earth System Model (CESM) workshops (Boulder, United States)
- Oral presentation at the 2025 ‘PDEs on the Sphere’ conference (São Paulo, Brazil).
- Oral presentation at the Parallel-in-Time (PinT) 2023 workshop (Hamburg, Germany)
- Oral presentation at the 2023 ‘PDEs on the Sphere’ conference (Grenoble, France)

- **Andrews, T.**, Wingate, B., and Shipton, J.: Quantifying the accuracy of large time-steps in highly oscillatory systems, EGU General Assembly 2023, Vienna, Austria, 23–28 Apr 2023, EGU23-14086, <https://doi.org/10.5194/egusphere-egu23-14086>, 2023.
- Visiting seminar at the National Institute of Water and Atmospheric Research (NIWA), 2023 (Wellington, New Zealand)

Scientific

Peer-Reviewed Articles

- **Andrews, T. C.**, Jablonowski C. (2026). ‘Stability analyses of divergence and vorticity damping on gnomonic cubed-sphere grids’. *Monthly Weather Review*, 154(7), 1317–1336.
- **Andrews, T. C.**, Wingate, B. A. (2026). ‘A numerical study of mean corrected phase-averaging in geophysical flow equations’. Accepted to *Theoretical and Computational Fluid Dynamics* (August 2026).

Manuscripts in review

- **Andrews T. C.**, Jablonowski C., and Hughes O. K., and Bendall T. M. (2026), ‘A mountain-generated mesoscale test case from DCMIP-2025: Gap flow and vortex shedding variants’. Submitted to *Geophysical Model Development*, available online at <https://egusphere.copernicus.org/preprints/2026/egusphere-2026-2293>.
- **Andrews, T. C.**, Bendall, T. M. (2026) ‘A conservative, discontinuous Galerkin, tracer transport scheme using compatible finite elements.’ Submitted to the *Journal of Computational Physics* and available as an arXiv preprint at <https://arxiv.org/abs/2603.19075>.
- **Andrews T. C.**, Duru K., Lee D., (2026) ‘A perfectly matched layer for damping vertically propagating waves in the nonlinear, compressible Boussinesq equations’. Submitted to the *Journal of Computational Physics* and available as an ArXiv preprint at <https://arxiv.org/abs/2606.28303>.
- Pletzer, T., Pletzer, A., **Andrews, T. C.**, Lin, D., and Katurji, M.: Mimetic interpolation for finite-time Lyapunov exponent computation near irregular domain boundaries, *EGUsphere* [preprint], <https://egusphere.copernicus.org/preprints/2026/egusphere-2026-3377>, 2026.

Preprints

- **Andrews, T. C.**, Shipton, J., Wingate, B. (2023), ‘The effect of linear dispersive errors on nonlinear timestepping accuracy in the f-plane rotating shallow water equations’. Available as an arXiv preprint at <https://arxiv.org/abs/2305.06685>.

In Preparation

- **Andrews, T. C.**, Jablonowski C., (2026), ‘A study of momentum equation diffusion in the CAM-FV3 dynamical core using baroclinic wave simulations’. In preparation for submission to *Geophysical Model Development*.

PhD Thesis

PhD Thesis from the University of Exeter: ‘Using nonlinear interactions to quantify and improve timestepping accuracy in the Rotating Shallow Water Equations’ (2024). Available at: <https://ore.exeter.ac.uk/articles/thesis/29811980?file=56858228>.

Other Research Experience and Skills

- A code developer for the Gusto, a compatible finite element library for solving the geophysical fluid equations (<https://www.firedrakeproject.org/gusto>).
- Connections with the UK Met Office, including code contributions to the trunk of LFRic’s shallow water mini app.

- Collaborations with Alex Pletzer at NIWA (New Zealand) for mimetic vector interpolation for LFRic, including comparing different Jacobian approximations.
- I can use the programming languages of Python, Fortran, MATLAB, R, and Visual Basic.
- Summer Internship research project at the Methanex New Zealand plant in Taranaki. This focused on using extreme value statistics to extrapolate the states of the plant's processing devices, given a small testing sample (2018-2019).
- Summer Research Scholarship project in the Precision Acoustic Group, within the University of Auckland Physics Department. I investigated using reflected acoustic measurements to estimate soil moisture properties (2017-2018).

Academic Service

- Volunteer peer reviewer for AMS Journals (2025).
- Instructor at the DCMIP-2025 summer school (<https://sites.google.com/umich.edu/dcmip-2025>) (2025).
- Judge for the Michigan Geophysical Union annual symposium (2025).
- Coordinator for scientific committee elections for the Parallel-in-time (PinT) community (2023, 2025).

Academic Achievements

During my bachelor's degree in Engineering Science:

2019

First in Course Awards for:

ENGSCI 711: Advanced Mathematical Modelling

ENGSCI 740: Advanced Mechanics in Research and Technology

ELECTENG 733: Signal Processing

2018

First in Course Awards for:

ENGSCI 314: Mathematical Modelling 3

ENGSCI 343: Mathematical and Computational Modelling in Mechanics

ENGSCI 344: Modelling and Simulation in Computational Mechanics

ENGSCI 391: Optimisation in Operations Research

MECHENG 325: Dynamics of Fluids and Structures

2017

First in Course Awards for:

ENGSCI 233: Computational Techniques and Computer Systems

ENGSCI 255: Modelling in Operations Research

2016 – 2018

Named on the University of Auckland Dean's Honours List for Top Engineering Students of the Year

Last year of High School:

2015

Dux of New Plymouth Boys' High School (NPBHS)

NZQA Scholarships attained in Physics, Statistics, and Economics

Awarded an IPENZ Foundation Scholarship to study Engineering

Teaching

- Guest lecturer for CLIMATE401: Geophysical FLuid Dynamics, University of Michigan (2025)
- Teaching assistant for NSC1002 Mathematics and Computing, University of Exeter (2022).
- High School maths and physics tutor for Taranaki Tutors, New Plymouth, New Zealand (2020–2022).

Other Achievements

- Ran the Wanganui marathon (42km) in under three and a half hours (2019)
- Have completed five half marathon runs
- Completed the ‘Lake Taupo Cycle Challenge’ (Approx. 160km) (2018)
- Member of the Auckland University Division 1 Men’s hockey team (2016-2017)
- Member of the 1st XI NPBHS Hockey team (2013-2015)
- Member of the NPBHS Senior Debating Team (2015)

Hobbies

I am passionate about outdoor activities such as hiking (we call it tramping in New Zealand!) and running. I love planning and executing multiple-day adventures in the forest parks of New Zealand. I am also aiming to complete the New Zealand Coast to Coast challenge sometime. In my spare time, when I’m not enjoying nature, I relax by cooking and reading epic fantasy books.

Academic Referees

Professor Beth Wingate - Department of Mathematics and Statistics, University of Exeter (Emeritus)
PhD supervisor
b.wingate@exeter.ac.uk

Professor Christiane Jablonowski - Department of Climate and Space Sciences and Engineering, University of Michigan
Postdoctoral research supervisor
cjablono@umich.edu